MIL-N-43181B
21 May 1986
SUPERSEDING
MIL-N-43181A
9 February 1970

MILITARY SPECIFICATION

NET, MULTIPURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of multipurpose net.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; General Specification For
MIL-C-43678	- Cord, Fibrous Polyester, Diamond Braid

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following document of the issue in effect on the date of invitation for bids or request for proposal, forms a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The net shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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* 3.2 Materials. (See 6.4.)

3.2.1 Yarn. The yarn shall be either 260 denier ± 5.0 percent, 3 ply, 3 turns per inch minimum in the ply, or 780 denier ± 5.0 percent, single yarn, 3 turns per inch minimum. The yarn shall be a bright, high tenacity, multi-filament nylon.

3.2.2 Cord, polyester, diamond braid. The cords for the suspension lines shall conform to the requirements of MIL-C-43678.

3.3 Color. The color of the finished net shall be Olive Green 106.

3.3.1 Color matching. The color shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7000 ± 500K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 100K.

3.3.2 Colorfastness. The finished net shall show fastness to light and laundering equal to or better than the standard sample. When a standard sample is not established or not designated as applicable to colorfastness, the net shall show good fastness to light and laundering when tested as specified in 4.2.2.3.

3.4 Physical requirements.

3.4.1 Knit. The netting shall be of the mesh type and shall be double faced and fully reversible.

3.4.1.1 Mesh size. The mesh size in the finished and relaxed state shall be 9/16 inch by 9/16 inch (inside measurements) ± 1/16 inch.

3.4.2 Weight. The weight of one finished net (see 3.4.3 and 3.7) shall be 16.5 ounces minimum (see 4.2.2.3).

3.4.3 Width. The width of the finished net inside selvages shall contain 100 ± 2 mesh repeats. The width of each selvage of the finished net shall be 1/4 inch minimum and 3/8 inch maximum.

3.4.4 Breaking strength of selvages. Each selvage of the finished net shall have a breaking strength of 185 pounds minimum when tested as specified in 4.2.2.3.

3.5 Heat setting. The material shall be heat set to stabilize the construction of the net.

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3.6 Soluble matter. The finished net shall contain not more than 2.0 percent of combined water soluble and chloroform soluble matter when tested as specified in 4.2.2.3.

3.7 Length of net. The length of the finished net shall be 108 inches minimum. The lengthwise meshes per foot shall be 10 ± 1 mesh. The count for the meshes per foot shall be made at the selvage, and the selvage shall be straightened by slight tension prior to count. In preparing the 108-inch lengths, the net shall be cut from selvage to selvage across the same course-wise line of holes by means of a hot knife or other suitable device which cuts and fuses the yarn. The cut edges shall have no loose ends. The lengths of the finished pieces shall be determined by measuring along the length of the selvage.

3.7.1 Length of cord. Two lengths of the cord specified in 3.2.2 shall be included and packaged with each net. Each length of cord shall be 14 feet (± 3 inches) long, and the ends of the cord shall be fused.

* 3.8 Workmanship. The finished net shall conform to the quality of product established by this document and shall be clean and free of holes, breaks, loose ends, and broken or missing yarns. The knit shall be of the specified mesh size. The net shall be of the specified size, color, and finish. The cord components shall have fused ends and be of the specified length. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of

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known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Visual examination. The completed nets shall be inspected for the defects in table I. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the net, in which case only one defect shall be counted. The lot size shall be expressed in units of one net each. The sample unit shall be one net complete with cords. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. Visual defects

Examine	Defect
Net	
Knit	Hole, break, weak place, mend
	Thin area
	Loose or floated end
	Coarse yarn more than twice the thickness of normal yarn
	Broken or missing yarn
	Loose knitting, causing sleaziness
Finished edge	Cut ends not fused

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TABLE I. Visual defects - Continued

Examine	Defect
Color	Off shade
Cleanness	Objectionable odor
	Overall uncleanness
Cord components	Cord cut, abraded or chafed
	Cord ends not fused
	Less than two lengths of cord per net

4.2.2.2 End item dimensional examination. The nets shall be examined for the measurement defects in table II. The lot size shall be expressed in units of one net each complete with cords. The sample unit shall be one net. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE II. Dimensional defects

Examine	Defect
Mesh	Mesh size not within tolerances
	Number of mesh repeats in width beyond tolerances
	Number of lengthwise meshes per foot less than 9 or more than 11
Net	Length of net less than minimum
Cord	Length of cord not as specified

4.2.2.3 End item testing. The finished net, complete with cords, shall be tested in accordance with the methods specified in table III and FED-STD-191, as applicable. The physical and chemical values specified in section 3 apply to the result(s) of the determination(s) made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be

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expressed in units of one net. The sample unit shall be one finished net. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (nets)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE III. Test methods

Characteristic	Reference paragraph	Test method
Material:		
Nylon identification	3.2.1	1/
Denier	3.2.1	1/
Tenacity	3.2.1	1/
Ply	3.2.1	Visual 2/
Turns per inch	3.2.1	Visual 2/
Luster	3.2.1	1/
Colorfastness to:		
Laundering	3.3.2	5614
Light	3.3.2	5660
Knit	3.4.1	Visual 2/
Weight	3.4.2	3/
Breaking strength, selvage	3.4.4	5100 4/
Finish:		
Heat set	3.5	1/
Water and chloroform soluble material	3.6	2611

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1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the results reported as "pass or fail."

3/ The weight of the sample unit shall be determined by a balance or scale capable of weighing the specimen accurately to the nearest 0.01 ounce.

4/ Specimens shall be cut from the net in such a manner that vertical pillars adjacent to the selvage remain uncut and there are no mesh repeats attached to the specimens. The long dimension of the jaws shall be no less than the width of the specimen, and the distance between the jaws at the start of the test shall be 10 inches.

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* 4.2.3 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete sealing or closure of container flaps, improper taping, loose strapping, or inadequate stapling Bulged or distorted container Open and noncontinuous heat-sealed seams (and closure) of polyethylene bags Omission of vent hole in polyethylene bag
Content	Number of intems per shipping container is more or less than required

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* 4.2.4 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each net shall be compactly rolled or folded. Each cord shall be coiled and secured by one of its loose ends or folded and securely tied with cord, tape, or twine. Each net, complete with two cords, shall be unit packed into a close-fitting flat or square style clear polyethylene film bag of 0.00125 inch thickness (± 20 percent). The polyethylene bag shall be formed with heat-sealed seams that are straight, continuous, and parallel to each other and to the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. A 1/4-inch diameter hole shall be made at one corner of each polyethylene bag to allow the excess air to escape.

5.1.2 Commercial. Each net, complete with cords, shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing. Thirty-two, complete with cords and preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads. The inside dimensions of the shipping container shall approximate 23 1/2 by 19 1/2 by 10 inches. Approximate dimensions are furnished as a guide only.

5.2.2 Level B packing. Thirty-two, complete with cords and preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3. The inside dimensions of the shipping container shall approximate 23 1/2 by 19 1/2 by 10 inches. Approximate dimensions are furnished as a guide only.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.3.

5.2.3 Commercial packing. Nets, complete with cords and preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), nets, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 90 in accordance with the appendix of MIL-STD-147.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall be marked with the following information:

a. Stock number
b. Item description
c. Quantity

The markings shall be clear and legibly printed or stamped in black. They will be placed directly on the bag across the center face or on a white paper label inserted inside the bag so as to permit ready identification.

6. NOTES

6.1 Intended use. The net (complete with cords) is intended to meet multiple requirements of Special Warfare units. It is usable as a hammock, camouflage net, carrier for bulky loads, a litter, or as a net or seine for subsistence fishing and catching game.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
c. Type and class of unit load required (see 5.2.1).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

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* 6.5 Knitting machine. The following is for information only and is not a mandatory requirement (see 3.4.1):

The standard sample was knitted on a two bar (needle and guide) Raschel machine with approximately 13 needles on each needle bar per inch (26 gage). The following chain notation for netting structure, exclusive of the selvage, may be used as a guide. The guide bars are to be threaded 1 in and 1 out on both bars.

	One bar		Other bar
	2 - 0		4 - 6
	0 - 2		6 - 4
	2 - 4		4 - 2
	4 - 2		2 - 4
	2 - 0		4 - 6
	0 - 2		6 - 4
7 Repeats	2 - 4	7 Repeats	4 - 2
	4 - 2		2 - 4
	4 - 6		2 - 0
	6 - 4		0 - 2
	4 - 2		2 - 4
	2 - 4		4 - 2
	4 - 6		2 - 0
	6 - 4		0 - 2
7 Repeats	4 - 2	7 Repeats	2 - 4
	2 - 4		4 - 2

6.6 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-N-43181	Net, Multipurpose

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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